

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B.Tech Examination (IT)

December 2011

IT302: Design & Analysis of Algorithm

Date: 14.12.2011, Wednesday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1

- (a) Define Space complexity. Explain with the help of a suitable example. [2]
- (b) What do you mean by correct and efficient algorithm? [2]
- (c) Suppose we are comparing implementations of insertion sort and merge sort on the same machine. For input of size n , insertion sort runs in $8n^2$ steps, while merge sort runs in $64n \lg n$ steps. For which values of n does insertion sort beat merge sort? [3]

Q-2

- (a) Find total no primitive operations and growth rate of a given algorithm for best case and worst case. [5]

Do_Something (int A[], int n)

```
{  
    for i = 2 to n  
    {  
        index = A[i];  
        j = i - 1;  
        while (j > 0) and (A[j] > index)  
        {  
            A[j+1] = A[j];  
            j = j - 1;  
        }  
        A[j+1] = index;  
    }  
}
```

- (b) Explain Greedy-choice property and Optimal substructure property by an example. [4]
- (c) Show that the running time of quick sort is n^2 when the array A is sorted in descending order. [5]

OR

Q-2

- (a) Give similarities and differences between backtracking and branch-and-bound [4]
- (b) Check the following equalities. [4]
 1. $n^2 = \Omega(n^3)$
 2. $2^n = O(2^{n+1})$
- (c) The sum of subset problem is to find all combination of the n given distinct numbers whose sum is M . Draw the state space tree for the problem with the numbers $S = \{3, 4, 5, 6\}$ and $M = 9$. [6]

Q-3

[14]

- (a) We have set of n jobs to execute, each of which takes unit time. At any time $T=1, 2$, we can execute one job. Job i earn us a profit $P_i > 0$ if and only if it is executed no late than time d_i (deadline). Develop a greedy algorithm to solve the above problem. Run the algorithm for $n=4$ and the following values.

i	1	2	3	3
p_i	50	10	15	30
d_i	2	1	1	7

- (b) Define the "n queen problem". Write an algorithm using backtracking to solve "n queen problem" and using the algorithm find the solution for 4 queen problem.

OR

Q-3

[14]

- (a) Sort the given numbers using merge sort: 85,24,63,45,17,31,96,50,27,7,90 Also determine asymptotic cost of Divide, Conquer and Combine using recurrence tree method.
- (b) What is the significance of asymptotic notations in algorithm analysis? Explain various asymptotic notations used in analysis of algorithm in detail.

SECTION-II

Q-4

- (a) What are the advantages of greedy approach over dynamic programming? [2]
- (b) Show the comparisons the naïve string matcher makes for the pattern $P=0001$ in the text $T=000010001010001$. [2]
- (c) Solve the following task assignment problem instance for minimization. [3]

	1	2	3
a	4	7	3
b	2	6	1
c	3	9	4

Q-5

- (a) Is dynamic programming a Top-Down or a Bottom-Up technique? Why? Explain with an example of making change problem. [5]
- (b) What is polynomial time reducibility? Give example. [3]
- (c) Solve the following recurrence exactly. [6]
- $$t_n = \begin{cases} n & \text{if } n=0,1 \text{ or } 2 \\ 5t_{n-1} - 8t_{n-2} + 4t_{n-3} & \text{otherwise} \end{cases}$$

OR

Q-5

- (a) Determine the Longest Common Subsequence of the given two string : $S1 : \text{ALGORITHM}$ [5]
 $S2 : \text{LOGERITHM}$. Show all the steps.
- (b) Differentiate between Deterministic Algorithms and Non Deterministic Algorithms. [3]
- (c) How many valid match and spurious hits does the Rabin-Karp matcher encounter in the text $T= 2359023141526739921$ when looking for the pattern = 31415. Take value of modulo $q=13$. [6]

Q-6

[14]

- (a) For the following four matrices find the order of parenthesization for the optimal chain multiplication.

$A_1 = 15 \times 5$, $A_2 = 5 \times 10$, $A_3 = 10 \times 20$, $A_4 = 20 \times 25$.

- (b) State Master Theorem and Solve following recurrence using Master Theorem.

1. $T(n) = 9T(n/2) + n$

2. $T(n) = T(2n/3) + 1$

OR

Q-6

[14]

- (a) Consider $n=3$, $(w_1, w_2, w_3) = (2, 3, 3)$ and $(p_1, p_2, p_3) = (1, 2, 4)$ and $m=6$ find the optimal solution using dynamic programming for 0/1 knapsack problem using given data.

- (b) What is Minimum Spanning Tree? Write down applications of minimum spanning tree. Find minimum spanning tree for a given graph using prim's algorithm.


